

Near-ultraviolet resonant-cavity light-emitting diode grown on Si

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ABSTRACT

Near-ultraviolet (NUV) light sources offer a range of notable advantages in industrial, scientific, and medical fields. In this study, we successfully fabricated AlGaIn-based NUV resonant-cavity light-emitting diodes (RCLEDs) directly grown on Si substrates and designed for operation at 369.5 nm. A nanoporous AlGaIn distributed Bragg reflector (DBR) was successfully fabricated by electrochemical etching as the bottom reflector, demonstrating a central wavelength of 370 nm and a peak reflectance as high as 99%. The influence of the resonant cavity effect on device performance was investigated by varying the number of the deposited dielectric DBR pairs as the top reflector. Under electrical injection, the device with seven pairs of SiO₂/Ta₂O₅ dielectric DBR demonstrated single-longitudinal-mode emission at a peak wavelength of 369.8 nm, with a full width at half maximum of approximately 1.8 nm, exhibiting exceptionally high spectral purity. The NUV RCLED developed in this study provides valuable insights for the development and application of next-generation NUV light sources.

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III-nitride near-ultraviolet (NUV) light sources possess sufficient energy to initiate photophysical and photochemical processes (for fluorescence excitation and curing) while avoiding the excessive destructiveness of far-ultraviolet light (UVC).^{1–5} This makes them indispensable in numerous high-tech and industrial fields, such as industrial solidification, sterilization and disinfection, biosensing, and ytterbium (Yb⁺) atomic clocks.^{6–9} Compared to other substrates, Si substrates have the advantages of larger wafer size and lower cost together with automated process lines, and hence, (Al)GaIn-on-Si technology may enable the fabrication of high-performance NUV light sources that are mass-producible and low-cost.^{10–12}

The incorporation of distributed Bragg reflectors (DBRs) on both sides of the active region to form an optical microcavity enables modulated light, leading to effectively narrowed emission linewidth and improved directivity.^{13,14} In the optical microcavity, the light usually emits from only one side with the reflectivity tailored by the deposited dielectric DBR, while the other side from which light does not exit typically requires a very high reflectivity. Commonly used mirrors for high reflectivity include metal mirrors, epitaxial III-nitride DBRs, and

dielectric DBRs.^{15–17} The technical approaches of using metal mirrors or dielectric DBRs require wafer bonding and substrate removal technologies, which add extra cost and complexity, whereas direct epitaxy of III-nitride DBRs is a much more competitive solution. Conventional AlInN/GaN or AlGaIn DBRs exhibit a small refractive index contrast, typically requiring about more than 40 pairs to achieve high reflectance ($\geq 95\%$). For instance, a 45-pair Al_{0.3}Ga_{0.7}N/Al_{0.04}Ga_{0.96}N DBR demonstrates a peak reflectance of only 95.4% at 368 nm.¹⁸ Such a large number of periods not only fails to deliver a high reflectivity with a large stopband but also presents a major challenge to the cost and overall material quality of the epitaxial DBR.

In recent years, several groups have employed electrochemically prepared porous GaN for photonic devices.^{19–22} Compared to other DBRs, the nanoporous (NP)-DBR structure offers a simpler and more stable fabrication process. To fabricate NP-DBR in the NUV region is more challenging due to the possible absorption of GaN as compared with the visible spectrum, and thus NP-AlGaIn DBR is much more preferred. However, reports on NP-AlGaIn DBR in the NUV band still remain limited.

This study successfully designed and fabricated an NUV resonant-cavity light-emitting diode (RCLED) that operated at approximately 369.5 nm. To achieve a high reflectivity at the bottom side, we employed electrochemical etching to create an NP-AlGaIn DBR structure on a Si substrate, demonstrating a peak reflectance as high as 99% at 370 nm. The modulation effect of the optical cavity on the NUV light source was systematically studied by comparing devices with different pairs of top dielectric DBRs. As a result, the NUV RCLED with seven-pair top dielectric DBR exhibited a single-mode emission at 369.8 nm with a narrow linewidth below 2 nm, showing significantly better spectral purity than conventional NUV LEDs. As far as we know, this is the first report of an NUV RCLED with a nanoporous AlGaIn DBR fabricated on Si, showing that this technology is fully compatible with AlGaIn on Si systems.

Figure 1(a) shows the epitaxial structure of the NUV RCLED directly grown on the Si (111) substrate by metal-organic chemical vapor deposition (MOCVD). It includes an AlN/AlGaIn buffer layer, a 1.3- μm -thick undoped GaN layer, a 1- μm -thick n-GaN layer, 20 pairs of alternately doped n-Al_{0.05}Ga_{0.95}N layers ([Si]: 1×10^{18} , $3 \times 10^{19} \text{ cm}^{-3}$) to fabricate NP-AlGaIn DBR, a 550-nm-thick n-Al_{0.05}Ga_{0.95}N layer, and an Al_{0.1}Ga_{0.9}N pre-layer, five pairs of In_{0.05}Ga_{0.95}N/GaN multiple quantum wells (MQWs), a p-Al_{0.25}Ga_{0.75}N electron blocking layer (EBL), a p-Al_{0.05}Ga_{0.95}N layer, and an about 10-nm-thick p⁺-GaN contact layer. The cavity length was designed to be 5.5λ .

Figures 1(b)–1(d) show a schematic diagram of the as-fabricated device structure. A comparative study was conducted on the modulation effects of the resonant cavity on the NUV light source by controlling the reflectivity of the top dielectric DBR. Three samples with the same highly reflective NP-AlGaIn DBR but different pairs of top SiO₂/Ta₂O₅ dielectric DBRs were fabricated and compared. Samples A, B, and C contained zero, four, and seven pairs of SiO₂/Ta₂O₅ dielectric DBRs, respectively.

The AlGaIn-based NUV LED epitaxial wafer, after a rapid thermal annealing at 550 °C in air for 30 s to activate the Mg acceptors, was processed into NUV RCLED through a series of processes. The sidewalls of the 20-period alternately doped n-AlGaIn layers were exposed by dry etching prior to electrochemical etching in a 0.3 M nitric acid solution under a forward bias of 5 V to form NP-AlGaIn DBR. Then, the current aperture was defined by dry etching with a depth of approximately 15 nm, followed by a deposition of 15-nm-thick SiO₂ by using inductively coupled plasma chemical vapor deposition (ICPCVD). Subsequently, a 30-nm-thick ITO layer was sputtered as the current spreading layer, followed by Ohmic contact annealing (525 °C in air for 3 min). The n-electrode mesa was defined by photolithography and dry etching. Then, the p- and n-electrodes were formed by evaporating Ti/Pt/Au (30/50/150 nm). Samples B and C were further fabricated with the deposition of a spacer layer, as well as four and seven pairs of SiO₂/Ta₂O₅ dielectric DBRs, respectively. As a result, the NUV RCLEDs were fabricated.

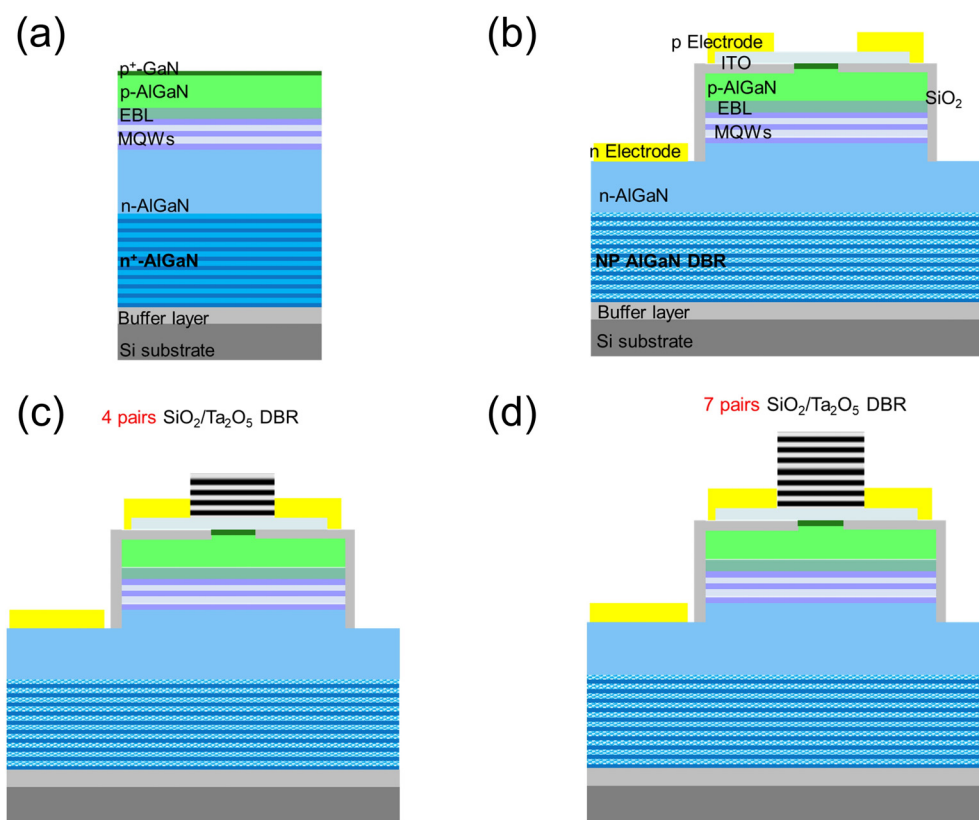


FIG. 1. Schematic diagrams of the NUV RCLED epitaxial and device structures. (a) Epitaxial structure. (b) Sample A with only NP-AlGaIn DBR. (c) Sample B with NP-AlGaIn and four-pair SiO₂/Ta₂O₅ dielectric DBRs. (d) Sample C with NP-AlGaIn and seven-pair SiO₂/Ta₂O₅ dielectric DBRs.

The crystalline quality of the AlGa_N film grown on Si was evaluated through double crystal x-ray rocking curve measurements obtained by high-resolution x-ray diffraction (Bruker D8 Discover), as presented in Fig. 2(a). The full width at half maximum (FWHM) of the AlGa_N film for the (0002) and (10 $\bar{1}$ 2) planes was 342 and 338 arc sec, respectively, corresponding to an estimated threading dislocation density (TDD) of $8 \times 10^8 \text{ cm}^{-2}$. Figures 2(b) and 2(c) show the cross-sectional scanning transmission electron microscopy (STEM) image of the as-fabricated device structure (sample C), demonstrating sharp interfaces of the AlGa_N-on-Si RCLED epitaxial materials.

A series of measurements of the as-fabricated NUV RCLEDs were performed at room temperature. These devices were tested directly on the chip and were not specially packaged. The cross-sectional image of the device was obtained by focused ion beam (FEI Scios) and STEM (FEI Tecnai G2 F20 S-Twin). The cross-sectional image for porosity calculation was obtained using a scanning electron microscope (SEM 4800). The reflectivities of NP-AlGa_N and SiO₂/Ta₂O₅ dielectric DBRs were measured by micro-spectrometer (Otsuka Electronics OPTM). The current-voltage (I-V) characteristic was measured by a direct-current power supply (Keithley 2450). The luminous image was taken with a CCD camera. The electroluminescence (EL) spectra and light output power were collected under continuous-current (CW) injection by using a spectrometer (Ocean Optics HR

4000) and a power meter (Thorlabs PM100D), respectively. The transmittance and absorption loss of ITO were measured by using a spectrophotometer (LAMBDA 750) and an ellipsometer (M-2000DI), respectively.

The NP-AlGa_N DBR as the bottom reflector was fabricated by directly growing alternating n-type doped layers and using electrochemical etching, thereby simplifying the process by avoiding substrate removal and wafer bonding processes, while maintaining compatibility with standard UV LED technology. Figure 3(a) shows the cross-sectional SEM image of the NP-AlGa_N DBR fabricated by electrochemical etching. The image reveals that the pore size and distribution were relatively uniform. Figure 3(b) presents the porosity calculation for the AlGa_N nanoporous structure. Based on data analysis, the porosity of the NP-AlGa_N layer was estimated to be approximately 34%, which is consistent with the results obtained from other areas (not shown here), indicating a good overall uniformity.

Based on the reported refractive indices of Al_{0.1}Ga_{0.9}N and GaN, the refractive index of n-Al_{0.05}Ga_{0.95}N is assigned as 2.7 at 369.5 nm in this study.²³ By using the volume average theory and the refractive index of the as-grown n-Al_{0.05}Ga_{0.95}N layer, the effective refractive index of NP-AlGa_N could be calculated to be approximately 2.28,²⁴ and hence the 20-period NP-AlGa_N DBR could yield a theoretical reflectance of about 99.3%. The practical reflectance of the NP-AlGa_N DBR was measured by using a micro-reflectance setup with a spot size

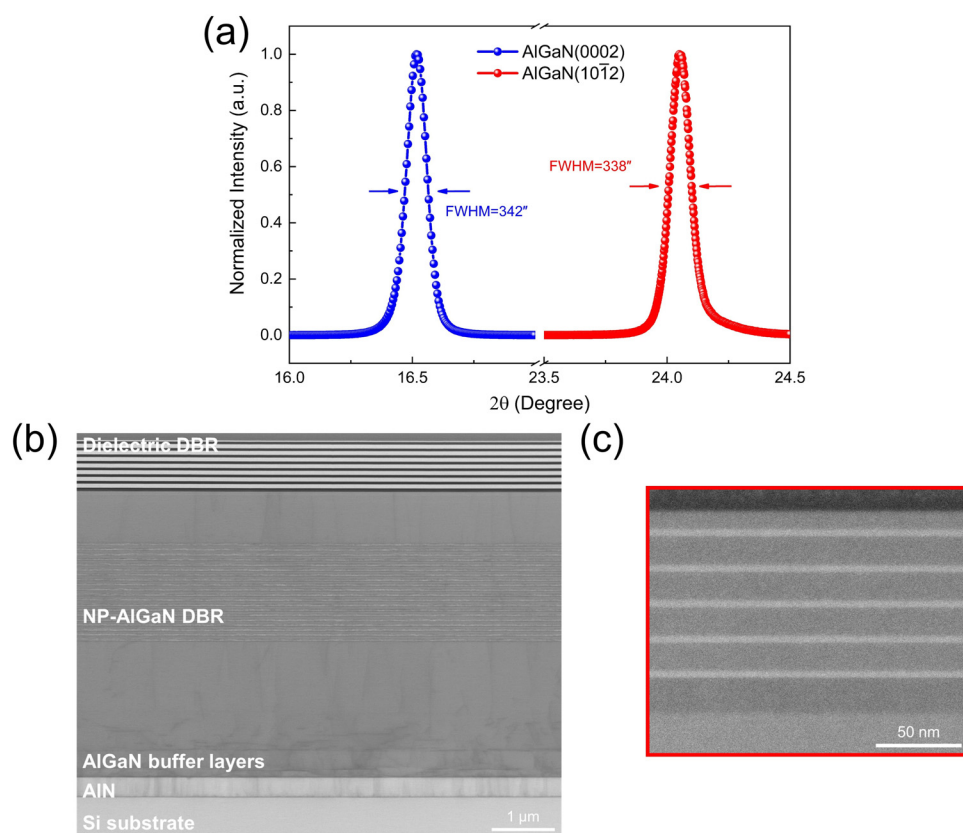


FIG. 2. (a) Double crystal x-ray rocking curves for the AlGa_N (0002) and (10 $\bar{1}$ 2) planes of the NUV structure grown on Si. (b) Cross-sectional STEM image of the as-fabricated device sample C. (c) Enlarged image of the MQWs active region.

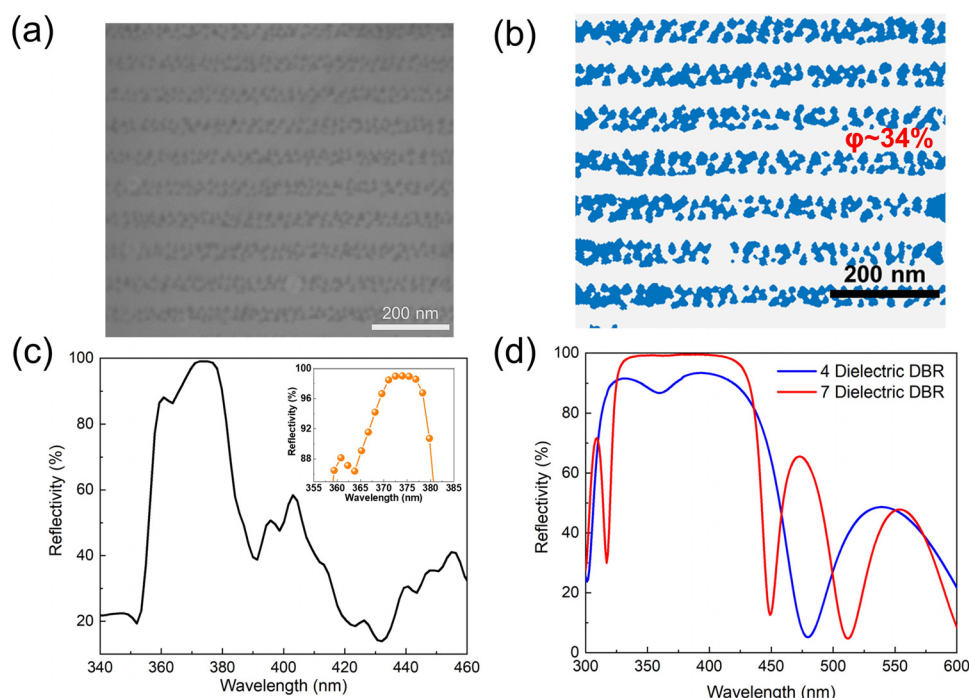


FIG. 3. (a) Cross-sectional SEM image of NP-AlGaIn DBR structure. (b) Calculation of porosity for the NP-AlGaIn structure. (c) Reflection spectrum of sample A and the inset displaying the peak reflectance value. (d) Reflection spectra of the four- and seven-pair $\text{SiO}_2/\text{Ta}_2\text{O}_5$ dielectric DBRs with a spacer layer (on Si wafer).

of $10\ \mu\text{m}$. Figure 3(c) displays the reflectance spectrum of NP-AlGaIn DBR, demonstrating a peak reflectance of $\sim 99\%$ at a central wavelength of $370\ \text{nm}$, as shown in the inset of Fig. 3(c). It is consistent with the theoretical value derived from the SEM image, showing good control of the nanopore size and uniform distribution and enabling seamless integration with the conventional fabrication processes of AlGaIn-based NUV light sources.

To compare the influence of various resonant cavity designs on the NUV light source, we first deposited a spacer layer, followed by four- and seven-pair $\text{SiO}_2/\text{Ta}_2\text{O}_5$ dielectric DBRs on the as-fabricated AlGaIn-on-Si wafers, which are named samples B and C, respectively. The reflectance measurements showed that the four- and seven-pair dielectric DBRs reached approximately 90% and 99% at about $370\ \text{nm}$, respectively, as shown in Fig. 3(d). Afterwards, we analyzed the device characteristics of the as-fabricated AlGaIn-on-Si RCLEDs. As illustrated in Fig. 4(a), sample A with a $12\text{-}\mu\text{m}$ -diameter electrical injection aperture (the red circle) was examined under a $3\ \text{mA}$ current injection, corresponding to a current density of approximately $2.7\ \text{kA}/\text{cm}^2$. The resulting light spot exhibited a relatively uniform emission intensity distribution within the current injection aperture, suggesting homogeneous current injection.

Figures 4(b)–4(d) present the emission spectra under CW current injection for sample A without any dielectric DBR, sample B with a four-pair dielectric DBR, and sample C with a seven-pair dielectric DBR, respectively. The EL spectrum of sample A exhibited a peak at approximately $365\ \text{nm}$, corresponding to the dip in the reflectance spectrum of sample A as shown in Fig. 3(c), which is mainly due to the modulation effect of the optical resonant cavity formed by the bottom

NP-AlGaIn DBR and the top reflection of the p-GaN/air interface. The FWHM of the EL spectrum narrows progressively with improved reflectivity of the top DBR and cavity quality, as evidenced by the values of 4.5 , 2.5 , and $1.8\ \text{nm}$ for samples A, B, and C, respectively. It demonstrates that the NUV light source can be engineered through various cavity designs, and a configuration with high reflectivity ($\sim 99\%$) on both facets results in a quite narrow linewidth and high spectral purity. All the devices show single-mode output due to a special cavity design. With an effective cavity length of $1\ \mu\text{m}$ and a mode spacing of about $25\ \text{nm}$, only one mode exists within the gain bandwidth, ensuring a good monochromaticity.

Figure 5 shows the output power–current (P–I) characteristics of the as-fabricated devices. It is evident that the light output power of sample A with only the NP-AlGaIn DBR was significantly higher than that of the devices with an additional dielectric DBR. At $10\ \text{mA}$, the external quantum efficiency of samples B and C is calculated to be 35% and 7.8% of sample A, respectively, as shown in Table I. This is an expected phenomenon, primarily attributed to the mirror loss caused by the high-reflectivity DBR and material absorption. Due to the highly reflective dielectric DBR, only a small portion of light could travel through and emit, and a lot of light was reflected back to the resonant cavity. Therefore, the resonant cavity effect is more pronounced and the output power is reduced. It was consistent with the reduced FWHM of the EL spectra and spot brightness.

Table II summarizes the performance of AlGaIn-based UV RCLEDs incorporating DBR structures, including the pair and reflectivity of epitaxial DBRs, emission wavelength, and FWHM. Compared with directly grown epitaxial DBRs, nanoporous DBRs are easier to

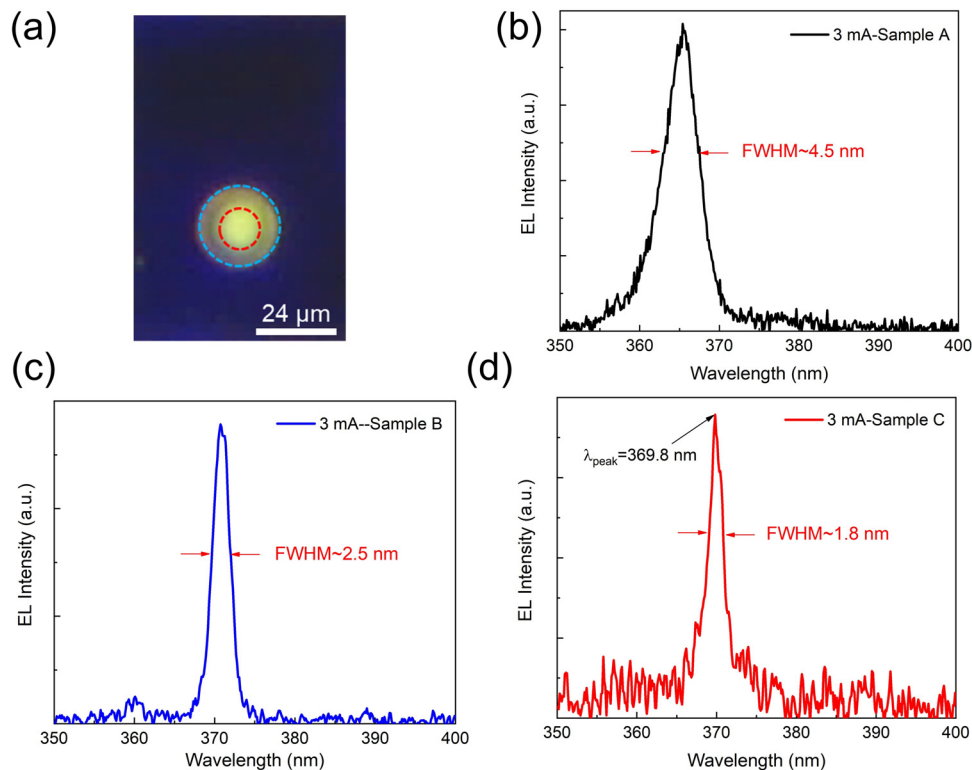


FIG. 4. (a) Top-view optical microscope image of sample A with a 12- μm -diameter current injection aperture at CW 3 mA. The red and blue circles show the sizes of the current aperture and the electrode aperture, respectively. EL spectra of (b) sample A, (c) sample B, and (d) sample C at 3 mA current injection, respectively.

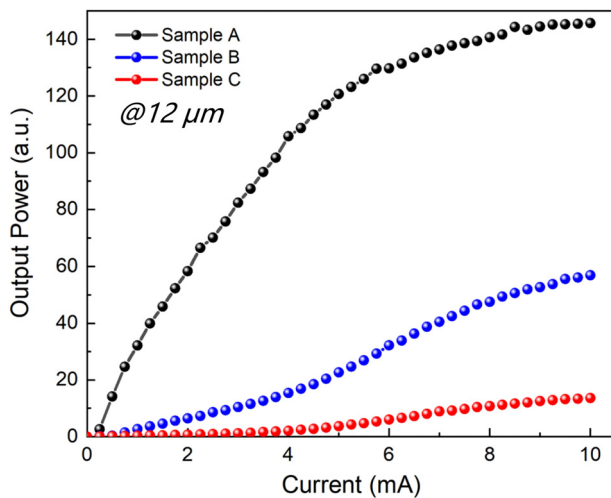


FIG. 5. Comparison of the P-I characteristics of samples A-C.

fabricate and more capable of achieving high reflectivity with even fewer pairs. To the best of our knowledge, this is the first report of such a high reflectivity achieved on a Si substrate in the NUV band, which further demonstrates the feasibility of this technology on a low-cost and large-area AlGaIn on Si platform.

A vertical-cavity surface-emitting laser (VCSEL) emitting at 369.5 nm exhibits excellent wavelength stability and a narrow spectral linewidth, making it a promising candidate as a light source for miniaturized Yb⁺ atomic clocks. So far, we have fabricated the device (sample C) with very high reflectivity ($\sim 99\%$) for both the bottom and top DBRs, targeted for NUV VCSEL. Figure 4(d) shows that the EL peak wavelength of sample C is approximately 369.8 nm. Figure 6(a) presents the P-I-V characteristics of sample C with a 12- μm current aperture under CW operations. The optical output power increases with the injection current density, and there is a clear turning point at around 3.5 kA/cm² in slope, indicating that the device operation enters a superluminescent state. When the current density further increased to around 9 kA/cm², the device experienced thermal failure due to the high junction temperature. Therefore, lasing behavior of the as-fabricated AlGaIn-based VCSEL structure grown on a Si substrate was not observed yet.

Several factors account for the non-lasing of the AlGaIn-based VCSEL. One of them is mainly due to the strong absorption loss of the ITO conductive layer. As shown in Fig. 6(b), the transmittance of the 30-nm-thick ITO layer was only 87% at a wavelength of 370 nm, corresponding to a very high optical absorption coefficient of about 6000 cm⁻¹, which is three to four times larger than that for blue wavelength.^{31,32} The ITO deposition parameters and post-deposition annealing conditions must be further optimized to minimize the optical loss in the ITO layer. Another reason is the high threading dislocation density, and the threading dislocations act as non-radiative

TABLE I. Summary and comparison of device performances for the three samples.

Sample	R _{Top} (%)	R _{Bottom} (%)	Cavity length	Wavelength (nm)	FWHM (nm)	EQE/EQE _{sample A} (%)
A	18			365.4	4.5	100
B	~90	~99	~5.5 λ	370.7	2.5	35
C	~99			369.8	1.8	7.8

TABLE II. Comparison of the performance of AlGaIn-based UV RCLEDs incorporating epitaxial-type DBR structures.

Sub.	DBR type	DBR pairs	R _{DBR} (%)	λ (nm)	FWHM (nm)	Ref.
Sapphire	Al _{0.3} GaN/Al _{0.04} GaN	45	~95.4	368	...	18
Sapphire	Al _{0.27} GaN/GaN	20	~90	390	...	25
Sapphire	Al _{0.12} GaN/GaN	40.5	~91.6	371.4	5.8	26
Sapphire	NP-AlGaIn/AlGaIn	20	~90	267.8	6.8	27
Sapphire	Airgap/AlGaIn	5	...	375	9.3	28
GaN	NP-AlGaIn/GaN	20	~99.5	370.5	2	29
Si	AlN/GaN	...	~96	380	...	30
Si	NP-AlGaIn/GaN	20	~99	369.8	1.8	This work

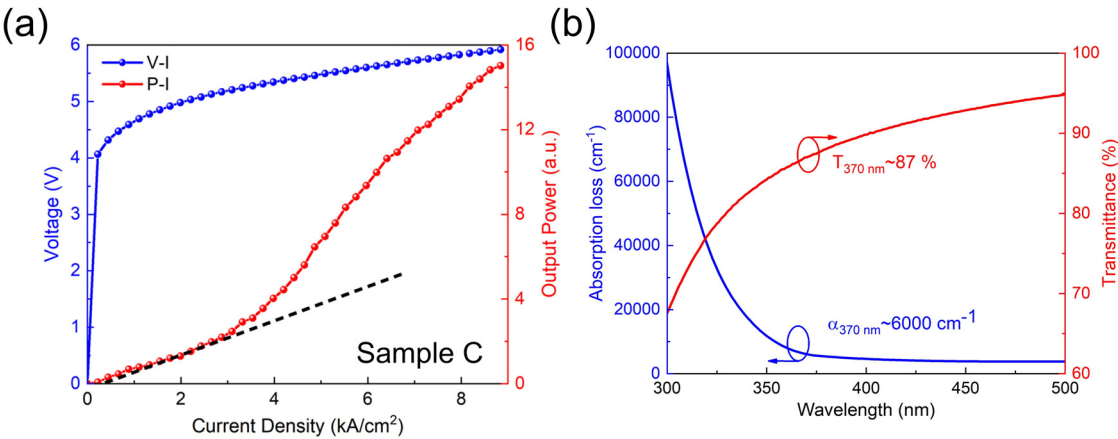


FIG. 6. (a) P–I–V characteristic of sample C with a 12- μ m current aperture under CW current injection. (b) Light absorption loss and transmittance of the 30-nm-thick ITO layer.

recombination centers, reducing the radiative recombination efficiency in the active region. Further optimization in ITO deposition, together with structural design and material growth, is still on the way to reduce the absorption loss and increase the emission efficiency, finally improving the device performance. These improvements pave the way for electrically injected lasing of NUV VCSEL at 369.5 nm, which is urgently required for the optical pumping of Yb⁺ atomic clocks.

In this study, we fabricated AlGaIn-on-Si NUV RCLEDs with dielectric and NP-AlGaIn DBRs, and the NP-AlGaIn DBR showed a high reflectivity of ~99% with a central reflection wavelength of ~370 nm, which is quite suitable for NUV RCLEDs and VCSELs. Under an electrical injection, the as-fabricated device exhibited single-longitudinal-mode emission with a peak wavelength at 369.8 nm and an FWHM of approximately 1.8 nm, displaying excellent spectral purity. To further improve the device performance, future work should

focus on reducing the optical loss and increasing the emission efficiency. This work provides valuable insights for the development of ultraviolet surface-emitting light sources and paves the way toward realizing electrically injected NUV VCSELs.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Chuanjie Li: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Meixin Feng:** Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Visualization (equal); Writing – review & editing (equal). **Xiujian Sun:** Methodology (equal); Resources (equal). **Jianxun Liu:** Methodology (equal); Resources (equal). **Shuming Zhang:** Project administration (equal); Resources (equal); Supervision (equal); Visualization (equal). **Qian Sun:** Funding acquisition (equal); Project administration (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **Hui Yang:** Data curation (equal); Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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